

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED								
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section -D 1.1.11 Sheet 1 of 10								
1.0 SCOPE This specification covers the technical requirements and essential particulars for the supply, application and finishing of the complete thermal insulation for cold equipment, piping systems, air-conditioning ducts etc. for temperatures between ambient and (-) 80°C and also for dual temperatures (both hot and cold) service with hot temperature above ambient and up to 230°C. Unless specified otherwise in section C and/or appendix -2, the scope of supply of the CONTRACTOR shall include, but not be limited to, the following items: a) Insulation materials of all types as specified and required. b) Insulation adhesives, vapour barriers and finishing materials of all types as specified and required. c) Auxiliary materials such as binding and lacing wires, wire netting, bands, screws, pop rivets etc., as specified and required. d) Angles, clamps, lugs etc. for supporting insulation e) Weather hoods f) Any other material as may be required for making the insulation complete 2.0 CODES AND STANDARDS 2.1 The supply and application of thermal insulation and finishing covered under this specification shall comply with all currently applicable statutes, regulations and safety codes in the locality where it is to be applied. The materials and application shall also conform to the latest editions of the codes and standards listed. Nothing in this specification shall be construed to relieve the CONTRACTOR of this responsibility. <table> <tr> <td>IS 277</td> <td>Galvanised Steel Sheets (Plain and Corrugated)</td> </tr> <tr> <td>IS 737</td> <td>Wrought Aluminium and Aluminium Alloy Sheet and Strip for General Engineering Purposes</td> </tr> <tr> <td>IS 3144</td> <td>Mineral Wool Thermal Insulation Materials - Method of Test</td> </tr> <tr> <td>IS 3677</td> <td>Unbonded Rock and Slag Wool for Thermal Insulation</td> </tr> </table> ISSUE R1			IS 277	Galvanised Steel Sheets (Plain and Corrugated)	IS 737	Wrought Aluminium and Aluminium Alloy Sheet and Strip for General Engineering Purposes	IS 3144	Mineral Wool Thermal Insulation Materials - Method of Test	IS 3677	Unbonded Rock and Slag Wool for Thermal Insulation
IS 277	Galvanised Steel Sheets (Plain and Corrugated)									
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Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section - D 1.1.11 Sheet 2 of 10
IS 4671 Expanded Polystyrene for Thermal Insulation Purpose IS 8183 Bonded Mineral Wool IS 9842 Preformed Fibrous Pipe Insulation IS 14164 Industrial Application and Finishing of Thermal Insulation Materials at Temperatures above (-) 80°C and up to (+) 750°C BS 3927 Rigid Phenolic Foam (PF) for Thermal Insulation in the Form of Slabs and Profiled Sections BS 5608 Preformed Rigid Polyurethane (PUR) and Polyisocyanurate (PIR) Foams for Thermal Insulation of Pipework and Equipment BS 5970 Thermal Insulation of Pipework and Equipment (in the Temperature Range (-) 100°C to (+) 870°C) 3.0 MATERIALS 3.1 GENERAL 3.1.1 All materials shall be new and fresh, incombustible, fire retardant, rot-proof, non-hygroscopic, vermin proof, fungus proof, non-injurious to health, chemically inert, non-corrosive to steel and aluminium (even if soaked in water for extended periods) and shall be guaranteed to withstand continuously and without deterioration the minimum and maximum temperatures to which these shall be subjected to under the specified applications. 3.1.2 The insulation materials and any component of the finished insulation job shall not react chemically, singly or in combination, with water or moisture to form substances that are more actively corrosive to applied surface than water or moisture alone. 3.1.3 In order to protect the workers from the hazards of insulation materials, suitable protective gadgets shall be provided. Required safety precautions shall be taken during handling and application of insulation. 3.1.4 The insulation material shall be kept dry at all times during transport, storage and installation. Decking and covering tarpaulins alone are not adequate for any length of time and shall not be allowed except in extreme emergencies and only for short period. Stacking of insulation materials directly on ground shall not be done. No wet insulation shall be installed. If wet insulation is present, it shall be removed and replaced with new dry insulation. The insulation may be protected with plastic film but shall be vented to prevent sweating ISSUE R1		

Spec.No.:
552H-E-MEC-
HVA-DM-S-E-
0001

THERMAL INSULATION FOR COLD SURFACES

Section - D 1.1.11
Sheet 3 of 10

3.2 INSULATION MATERIAL

The insulation material shall be one or more of the following types as specified in appendix -2

SL. NO.	INSULATION MATERIAL	STANDARD	DENSITY , Kg/M ³	TEMP. RANGE, °C	MATERIAL CODE
1.	Fire-retardant quality expanded polystyrene	IS 4671	20	(-)80 to 80	EP
2.	Rigid polyurethane / Polyisocyanurate foam	BS 5608	32	(-)80 to 110	PUR/ PIR
3.	Phenolic foam	BS 3927	32	(-)80 to 130	PF
4.	Lightly resin-bonded glass wool mattresses	IS 8183	32	(-)40 to 230	LRG
5.	Resin-bonded glass wool pipe sections	IS 9842	85	(-)40 to 230	PSG
6.	Unbonded rock and slag wool	IS 3677	150	(-) 80 to 230	U150

3.3 INSULATION ADHESIVE

The insulation adhesive shall be one or more of the following types as specified in appendix -2

	Material Description	Material Code
3.3.1	Hot bitumen of grade 85/25 or 85/40 conforming to IS 702 shall be uniformly applied at 1.5 Kg/M ² on the surface to be insulated. A similar layer shall also be applied on the inside surface of the insulation. This material shall not be used for stainless steel surfaces.	BIT
3.3.2	CPRX compound shall be uniformly applied at 1.5 Kg/M ² on the surface to be insulated. A similar layer shall also be applied on the inside surface of the insulation. This material shall not be used for stainless steel surfaces.	CPRX
3.3.3	MAS 83 with 0.5 mm wet film thickness shall be uniformly applied at 0.5 lit/M ² by trowelling. MAS 83 shall not be used with polystyrene as insulation material. MAS 83 can be used only between (-) 46 and (+) 149°C.	MAS 83

ISSUE
R1

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED																								
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section - D 1.1.11 Sheet 4 of 10																								
3.4 VAPOUR BARRIERS. The vapour barrier shall be one or more of the following types as specified in appendix -2. <table border="0" data-bbox="236 622 1426 1294"> <thead> <tr> <th></th> <th>Material Description</th> <th><u>Material Code</u></th> </tr> </thead> <tbody> <tr> <td>3.4.1</td> <td>Hot bitumen of grade 85/25 or 85/40 conforming to IS 702 shall be uniformly applied at 2.5 Kg/M² on the outer surface of insulation and allowed to dry. Reinforced Plastic (RP) tissue paper shall be applied on the dried bitumen. This material shall not be used for stainless steel surfaces.</td> <td>VB1</td> </tr> <tr> <td>3.4.2</td> <td>0.1 mm thick factory-laminated aluminium foil with craft paper shall be provided. All joints shall be provided with 50 mm overlap and all joints shall be sealed with aluminium tape. For temperature between (-) 43 and (+) 149°C MAS 35 may also be used as sealant.</td> <td>ALF</td> </tr> <tr> <td>3.4.3</td> <td>MAS 130 with 1.02 mm wet film thickness primer coat at 1.0 lit/M² and 2.25 mm wet film thickness finish coat at 2.5 lit/M² shall be uniformly applied over the insulation by spatula or trowelling. MAS 130 shall not be used with polystyrene as insulation material. MAS 130 can be used only between (-) 40 and (+) 60°C.</td> <td>MAS 130</td> </tr> </tbody> </table> 3.5 FINISHING MATERIALS The finishing material shall be one or more of the following types as specified in appendix -2. <table border="0" data-bbox="236 1523 1426 1854"> <thead> <tr> <th></th> <th>Material Description</th> <th><u>Material Code</u></th> </tr> </thead> <tbody> <tr> <td>3.5.1</td> <td>Portland cement and inorganic fibre with water proofing compound at an application density of 1050 to 1100 Kg/M³.</td> <td>FC</td> </tr> <tr> <td>3.5.2</td> <td>Aluminium sheet as per IS 737, designation 31000, condition H3.</td> <td>AL1</td> </tr> <tr> <td>(a)</td> <td>Insulation outside diameter above 450 mm - sheet thickness 18 SWG</td> <td></td> </tr> </tbody> </table> ISSUE R1				Material Description	<u>Material Code</u>	3.4.1	Hot bitumen of grade 85/25 or 85/40 conforming to IS 702 shall be uniformly applied at 2.5 Kg/M ² on the outer surface of insulation and allowed to dry. Reinforced Plastic (RP) tissue paper shall be applied on the dried bitumen. This material shall not be used for stainless steel surfaces.	VB1	3.4.2	0.1 mm thick factory-laminated aluminium foil with craft paper shall be provided. All joints shall be provided with 50 mm overlap and all joints shall be sealed with aluminium tape. For temperature between (-) 43 and (+) 149°C MAS 35 may also be used as sealant.	ALF	3.4.3	MAS 130 with 1.02 mm wet film thickness primer coat at 1.0 lit/M ² and 2.25 mm wet film thickness finish coat at 2.5 lit/M ² shall be uniformly applied over the insulation by spatula or trowelling. MAS 130 shall not be used with polystyrene as insulation material. MAS 130 can be used only between (-) 40 and (+) 60°C.	MAS 130		Material Description	<u>Material Code</u>	3.5.1	Portland cement and inorganic fibre with water proofing compound at an application density of 1050 to 1100 Kg/M ³ .	FC	3.5.2	Aluminium sheet as per IS 737, designation 31000, condition H3.	AL1	(a)	Insulation outside diameter above 450 mm - sheet thickness 18 SWG	
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Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES Material Description	Section - D 1.1.11 Sheet 5 of 10 Material Code
	(b) Insulation outside diameter 150 to 450 mm - sheet thickness 20 SWG (c) Insulation outside diameter below 150 mm - sheet thickness 22 SWG (d) Air-conditioning ducts - sheet thickness 24 SWG (e) Equipment - sheet thickness 18 SWG 3.5.3 Galvanised Steel (GS) sheet as per IS 277, designation GP GS1 (a) Insulation outside diameter above 450 mm - sheet thickness 20 SWG (b) Insulation outside diameter 150 to 450 mm - sheet thickness 22 SWG (c) Insulation outside diameter below 150 mm - sheet thickness 24 SWG (d) Air-conditioning ducts - sheet thickness 26 SWG (e) Equipment- sheet thickness 20 SWG 3.5.4 MAS 134 with 1.0 mm wet film thickness at 1.0 lit/M² shall be uniformly applied when using over porous insulation materials. Wet film thickness shall be 0.5 mm at 0.5 lit/M² for impervious insulation surfaces including polystyrene. MAS 134 can be used only between (-)18 and (+) 82°C. MAS 134 3.6 AUXILIARY MATERIALS 3.6.1 Binding and Lacing Wires Binding and lacing wire shall be annealed GS 24 SWG. 3.6.2 Wire Netting Wire netting shall be with GS 24 SWG wire and 20 mm hexagonal opening. 3.6.3 Bands (a) For securing insulation material : 24 SWG GS, 20 mm wide (b) For securing aluminium and GS : 24 SWG anodised aluminium	GS1 MAS 134 ISSUE R1

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section - D 1.1.11 Sheet 7 of 10
4.1.5 For fluid operating temperatures below (-) 20°C, the insulation shall be applied directly over the surface to be insulated without applying the insulation adhesive. Vapour barrier shall be applied over the insulation material. Vapour barrier and insulation finish shall be applied as per paras 3.4 and 3.5 or 4.5 respectively. 4.1.6 Where multilayer insulation is provided, insulation adhesive shall be used between two layers, based on temperature criterion given in para 4.1.4. 4.1.7 If aluminium or GS sheet is specified as finishing material, all joints shall be sealed with bitumastic paint and made effectively weather and water-proof. For temperatures between (-) 40 and (+) 120°C, MAS 94 may also be used as sealant. MAS 94 shall, however, not be used with polystyrene as insulation material. All flat surfaces shall be adequately sloped to prevent pools of water collecting. The sheet shall be protected internally with 2 coats of bitumastic paint. 4.1.8 All actions shall be taken to complete the application of finishing on exposed surfaces covered with insulation before closing the day's work. If this is not practicable, adequate precautions shall be taken to protect the insulation from weather, for example by wrapping it with polythene sheet, roofing felt or other approved material. 4.2 PIPES, PIPE FITTINGS, VALVES AND SPECIALITIES 4.2.1 All vertical pipes shall be provided with suitable insulation supports to prevent the insulation from collapsing due to its own weight. Any welding required, shall be carried out by the CONTRACTOR with the prior permission of the EMPLOYER and only under his direct supervision. Where welding is not permitted, suitable clamped supports shall be used. The insulation shall be applied starting from bottom up. Mattress type insulation materials shall be clamped from top. 4.2.2 All pipe fittings, valves and specialties shall be covered with the same type and thickness of insulation as specified for the adjoining pipe. On pipe fittings, insulation outside diameter shall be same as the outside diameter of adjacent pipe insulation. Unless specified otherwise in section C, valves and specialties of sizes 100 mm NB and larger shall be provided with removable box type insulation. Box shall be fabricated from sheet material specified for adjoining pipes. Pipe insulation on adjoining flanges shall be stopped at one bolt length plus 25 mm before flange to permit removal of the bolts and nuts. The insulation shall be applied after the finish has been applied over insulation on the adjacent piping. Flanged joints shall also be insulated with removable type of boxes. Arrangement shall be similar to that for valves. 4.2.3 The sheets shall be installed with the longitudinal lap joints at 45° below the horizontal for horizontal pipes and the joints sealed with bitumastic paint.		
		ISSUE R1

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section - D 1.1.11 Sheet 8 of 10
4.2.4 On vertical pipes the sheets shall be applied working from bottom up. Each section of sheets shall have a minimum overlap of 50 mm longitudinally and circumferentially. Each circumferential joint shall be made weather-proof by securing with a band of sheet material and sealing with bitumastic paint. Longitudinal lap joints shall be fixed with screws or pop rivets at approximately 150 mm centres. 4.2.5 Valves shall be insulated up to and including their bonnet flange. 4.3 EQUIPMENT 4.3.1 Where the insulation material is in the mattress form, cleats in the form of wire nails or nuts or angles and flats for supporting the insulation material, shall be welded to the equipment by others. If wire nails are to be used as insulation cleats, these shall be bent and secured with the metal fabric of the mattress, after the insulation has been applied. Where insulation cleats are in the form of M6 and M10 nuts, the CONTRACTOR shall supply and install bolts of suitable length for fixing the insulation. The insulation applied to equipment shall be reinforced with wire netting. One course of wire netting shall be applied to the surface of the equipment and each layer of insulation shall be backed up with wire netting. All irregularities of the surface shall be filled and levelled over with insulating cement. All mattress joints shall be butted tightly and the mattresses shall be secured with 20 mm wide 24 SWG GS bands at 450 mm centres. After banding, all mattress edges shall be laced tightly. 4.3.2 All equipment, unless specified otherwise, shall have a smooth aluminium or GS sheet finish as specified in data sheet A, applied in a manner similar to that specified for piping. For fixing of aluminium or GS sheets, wooden spacer rings at 1000 mm centres shall be fixed to the equipment by the CONTRACTOR. All vertical and horizontal sheets shall be overlapped a minimum of 75 mm. The lapped joints of adjoining sections of sheets shall be secured with screws or pop rivets. On all equipment above 2500 mm diameter and flat surfaces, the sheet shall be further secured by circumferential bands at approximately 1000 mm centres. Each sheet joint shall be sealed with bitumastic paint. The roof sections shall overlap the side walls to prevent water seepage between insulation and the equipment wall. Side wall sheets shall be securely banded at intersections of the side wall and roof sections. 4.3.3 All equipment manholes, hatches, bolted or screwed cover plates, flanged ends etc. shall have removable box type insulation, with same thickness of insulation as for adjacent surfaces. Insulation adjoining such equipment openings shall be tapered towards these openings to permit removal of bolts, screws, heads, covers or plates with no damage to adjacent surface insulation or cover. 4.3.4 Nozzles and other connections on tanks and other equipment shall be insulated in the same manner as the pipes.		
		ISSUE R1

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section - D 1.1.11 Sheet 9 of 10
4.3.4 Pump casing shall be completely insulated with removable type of boxes fabricated from the specified sheet material. Proper care shall be taken to maintain continuity of vapour barrier between static and removable portion of insulation. 4.3.5 Name plates on equipment shall not be insulated. Proper care shall be taken to eliminate seepage of moisture from such un-insulated portions into the insulating material. 4.4 CONTRACTION JOINTS IN INSULATION Depending on the type of insulation material used and the operating temperature, contraction joints shall be provided for equipment or pipes to prevent rupturing or buckling when the cold surface contracts. 4.5 CEMENT SAND PLASTER FINISH Where cement sand plaster finish is specified following procedure shall be followed : 4.5.1 Indoor Application After the application of hot bitumen or CPRX compound vapour barrier, the surface shall be wrapped with GS wire netting, butting all the joints and lacing down with lacing wire. The ends of all wire loops shall be firmly twisted together with pliers, bent over and carefully pressed into the surface of insulation. A total of 15 mm thick finishing cement sand(1:4) plaster shall be applied in two layers after a GS wire netting is stretched over the entire surface of insulation material and securely fastened down. The plaster shall be hand- trowelled over the wire netting to a smooth finish. The application of finishing plaster shall be terminated, at the end of day's work, at a contraction joint. 4.5.2 Outdoor Application After the application of the plaster as per procedure for indoor application, an additional 6 mm thick water-proofing compound Shalikota 30 or equivalent shall be applied in two layers. Outdoors, the finishing plaster shall be applied immediately after application of insulation. If this is not practicable, the CONTRACTOR shall take adequate precautions to protect the insulation from weather, e.g. by wrapping it with polythene sheet, roofing felt or other approved material.		

ISSUE R1

 MAHAN ENERGEN LIMITED	2x800 MW (PH-II) MAHAN ULTRA SUPERCRITICAL THERMAL POWER PROJECT	 TATA CONSULTING ENGINEERS LIMITED
Spec.No.: 552H-E-MEC- HVA-DM-S-E- 0001	THERMAL INSULATION FOR COLD SURFACES	Section D 1.1.11 Sheet 10 of 10

4.5.3 The cement sand plaster finish shall have the following properties :

- a) A light shower of rain, falling immediately after application shall not wash off the plaster.
- b) At any time, one week or more after application, it shall not shatter if struck a sharp blow with a 0.7 Kg hammer. Such a blow may damage the finishing locally but shall not cause large pieces to break away.
- c) When set, it shall withstand prolonged exposure to the weather without additional protection.

5.0 DUAL TEMPERATURE INSULATION

For dual temperature application, resin-bonded glass wool or unbonded rock or slag wool thermal insulation shall be provided, depending on the lower temperature limit given in para 3.2. The insulation application procedure shall be same as that for cold surfaces.

6.0 MEASUREMENT

6.1 Measurement of insulation over equipment and piping shall be as per IS 14164.

6.2 Measurement for air-conditioning ducts shall be taken over finished insulation surface and 5% additional area shall be considered to account for reinforcement and flanged angle joints.

7.0 MISCELLANEOUS

Approval of the EMPLOYER shall be obtained of samples of all materials and necessary test certificates of approved national laboratories, before despatching these to site. Insulation shall not be applied until specific release is given by the EMPLOYER.

ISSUE R1

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